Diagnostic: six questions, 10 minutes

Answer without consulting the lesson. Mark the questions for which help is needed. This diagnostic is not graded.

1. What is the slope of the line through $(1,2)$ and $(3,8)$?
2. Expand $(2+h)^2 - 4$.
3. Simplify $\dfrac{4h + h^2}{h}$ if $h \neq 0$.
4. Calculate $\sqrt{0}$ and $\sqrt{0.01}$.
5. Calculate $|-3|$ and $|2|$.
6. A line of slope 2 passes through $(1,5)$. What is its equation?

After the diagnostic. Revisit the flagged topics with the teacher; redo a similar question, then explain the rule used.

Speaking and logic. Explain to a classmate why we simplify the rate before letting h tend to 0. Distinguish average speed from instantaneous speed.

Investigate before formalising

For $f(x) = x^2$, let h vary, positive and negative, around 0, keeping the point A with x-coordinate 1. Draw three secants; conjecture a limiting line, then determine it without reading a graph.

Organisation. Individual investigation 5 min, comparison in pairs 5 min, class discussion 5 min. Keep a record of one attempt, even an unfinished one. A conjecture must then be justified.